done by increasing the sample

## Test for uniformity

**Frequency Test:-**

→ There are two methods available to test the uniformity.

→ Those are Kolmogorov – Smirnov test and chi – square test.

→ Both measure the degree of agreement betw^n the distribution of sample of generated random numbers and theoretical uniform distribution.

## The Kolmogorov Smirnov Test:-

→ In statistics the K-S test is used to determine whether two underlying probability distributions differ or whether an underlying probability distribution differs from a hypothesized distribution in either case based on finite samples.

→ The K-S test compares the continuous CDF $F(x)$ of uniform distribution specified by null hypothesis with empirical CDF, $S_N(x)$ of sample of $N$ observations.

→ We know that

The continuous CDF of uniform distribution is

$F(x) = x, \quad 0 \leq x \leq 1$

The empirical CDF of sample random numbers $R_1, R_2 \ldots R_N$ is.

$$S_N(x) = \frac{\text{no of } R_1, R_2 \ldots R_N \text{ which are } \leq x}{N}$$

(3).

Candidate Roll No. ____________
(In figures)

Name: ____________

Date: ____________ 20

Examination: ____________ Branch/Semester ____________

Subject: ____________

Test Exam.

Junior Supervisor's full Signature with Date

| Question No. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | Total |
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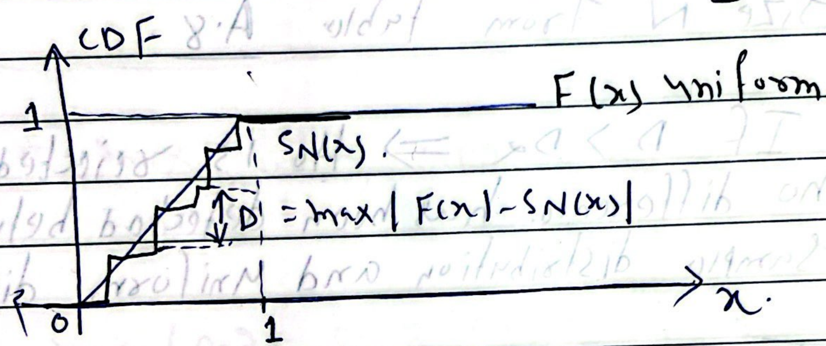
→ The test is based on largest absolute deviation between $F(x)$ and $S_N(x)$ over the range of random variable.

→ Plot $F(x)$ and $S_N(x)$ and get the statistic $D = \max|F(x) - S_N(x)|$ which is maximum vertical distance as shown in fig below.




→ The computed value of D is compared with tabulated value of sampling distribution (critical value) $D_\alpha$ for specified value of significance level '$\alpha$ and sample size N.

→ If compared value is greater than critical value then reject the null hypothesis. The value of D can be computed by using following algorithm also.

Algorithm

1) Define the hypothesis for testing the uniformity as.

$H_0: R_i \sim U[0,1]$.

$H_1: R_i \nsim U[0,1]$.

2) Arrange the data in increasing order.

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Let $R_{(i)}$ denote the ith smallest no then
$R_{(1)} \leq R_{(2)} \leq ---- \leq R_{(N)}$.

3) Compute $D^+$ and $D^-$ where:

$$D^+ = \max_{1 \leq i \leq N} \left\{ \frac{i}{N} - R_{(i)} \right\} \text{ and.}$$

$$D^- = \max_{1 \leq i \leq N} \left\{ R_{(i)} - \frac{i-1}{N} \right\}$$

4) ~~Compute $D^+$ and $D^-$ where.~~ Compute $D = \max(D^+$

5) Determine the critical value $D_\alpha$ for specified significance level $\alpha$ and the given sample size N from table A.8.

6) If $D > D_\alpha \Rightarrow H_0$ is rejected, else no difference has been detected betwn the sample distribution and uniform distribution.

→ The Kolmogorov - Smirnov test is more powerful and can be used for small sample sizes.

Example (June - 2011, May 2012)

The sequence of numbers 0.63, 0.49, 0.24, 0.89, 0.57 and 0.71 has been generated. Use the Kolmogorov Smirnov test with $\alpha = 0.05$ to determine if hypothesis that nos are uniformly distributed on the interval [0, 1] can be rejected.

⇒ Solution Method-1

1) Define the hypothesis for testing the uniformity as
$H_0: R_i \sim U[0,1]$.
$H_1: R_i \nsim U[0,1]$.

2) Rank data in increasing order.
$0.24 \le 0.49 \le 0.57 \le 0.63 \le 0.71 \le 0.89$.

3) Compute $D^+$ and $D^-$.

| $i$ | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| $R(i)$ | 0.24 | 0.49 | 0.57 | 0.63 | 0.71 | 0.89 |
| $i/N$ | 0.17 | 0.33 | 0.50 | 0.67 | 0.83 | 1.00 |
| $i/N - R(i)$ | – | – | – | 0.04 | 0.12 | 0.11 |
| $R(i) - \frac{(i-1)}{N}$ | 0.24 | 0.32 | 0.24 | 0.13 | 0.04 | 0.06 |

$D^+ = \max(0.04, 0.12, 0.11)$
$= \underline{0.12}$

and $D^- = \max(0.24, 0.32, 0.24, 0.13, 0.04, 0.06)$
$= \underline{0.32}$

4) Compute D
$D = \max(D^+, D^-)$.
$= \max(0.12, 0.32) = \underline{0.32}$

5) Determine the critical value $D_\alpha$ for specified level of significance $\alpha = 0.05$ and sample size $N = 6$.
$D_{0.05} = \underline{0.521}$ (obtained from table A.8)

6) Since $D = 0.32 < D_{0.05} = 0.521$
$\therefore$ $H_0$ is not rejected
From this we can say that given set of random numbers are uniformly distributed.

Method - 2

For getting the value of D, Plot the CDF of empirical distribution ($S_N(x)$) and uniform distribution ($F(x)$). The largest vertical distan[ce] between $F(x)$ and $S_N(x)$ is the D.

1) Define hypothesis for testing the uniformity a[s]

$H_0$ : $R_i \sim U[0,1]$.

$H_1$ : $R_i \nsim U[0,1]$

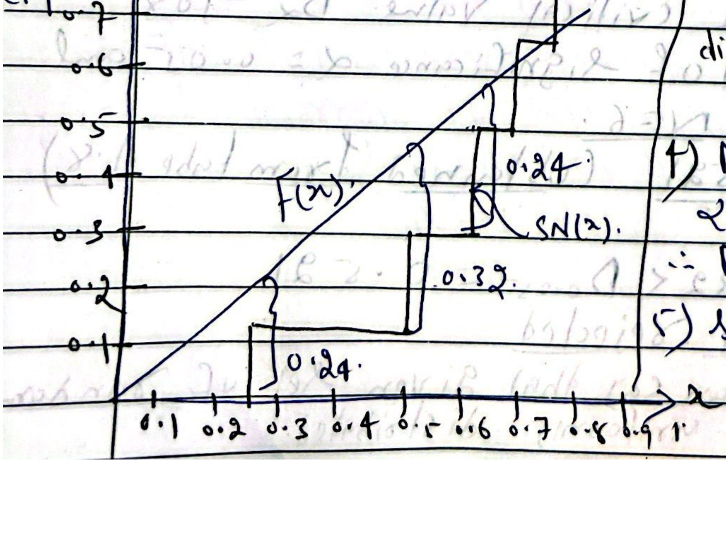
2) Compute the uniform CDF, ($F(x)$) and empirical CDF, ($S_N(x)$).

| X | 0.63 | 0.49 | 0.24 | 0.89 | 0.57 |
|---|---|---|---|---|---|
| $F(x) = x$ | 0.63 | 0.49 | 0.24 | 0.89 | 0.57 |

$S_N(x) = \frac{\text{no of } R_1, R_2 \ldots R_N \text{ which are} \leq x}{N}$

→ 0.67 0.33 0.17 1.00 0.50 0.[...]

3) Plot the CDF of empirical and uniform distribution




From F is largest vertica[l] distance is 0.32

D = 0.32

4) Determine $D_\alpha$ from tab[le]

$\alpha = 0.05$, N = 6.

$\therefore D_{0.05} = 0.521$

5) Since $D = 0.32 < D_{0.05} = 0.521$ ∴ $H_0$ is not rejected

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4

Candidate Roll No. ____________
(In figures)

Test Exam.

...e: ____________

...e: ____________ 20

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| Junior Supervisor's full Signature with Date |
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| ...uestion No. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | Total |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
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## * The chi-square Test:-

→ Pearson was first to recognize the practical value of chi-square test.

→ The chi-square test can be used to test the uniformity (or homogeneity) within a data set or to determine the probability that there is a dependency relationship betwn. two or more distinct data sets. (e.g in industry, the chi square test could be used to determine the probability that a particular machine in a group breaks down too often or that production increases when a raw material is changed).

→ This is: statistical ~~Prode~~ procedures whose results are evaluated by ref to chi-square distribution. It tests a null hypothesis that the relative frequencies of occurence of observed events follow a specified frequency distribution.

→ The problem is to decide whether N independent samples can be belonging to a common population. The algorithm for chi-square test is given below

1) Define the hypothesis for testing the uniformity as,

$H_0 : R_i \sim U[0,1]$

$H_1 : R_i \nsim U[0,1]$

2) Divide the total no of observations (N) into mutually exclusive equally numbered classes (n) [$(a_0, a_1)$ $(a_1, a_2)$ ---- $(a_{n-1}, a_n)$]. The n be chosen such a way that each $E_i \geq 5$.

3) Compute the sample test statistics.

$$\chi_0^2 = \sum_{i=1}^{n} \frac{(O_i - E_i)^2}{E_i}$$

Where.

$\chi_0^2 \rightarrow$ approximate chi square distribution with n- degrees of freedom.

$O_i \rightarrow$ observed number in ith class.

$E_i \rightarrow$ expected number in ith class which is N

4) Determine the critical value for specified significance level $\alpha$ with n-1 degrees of freedom from table A.6

5) If $\chi_0^2 > \chi_{\alpha, n-1}^2 \Rightarrow$ Ho is rejected ;

else no difference has been detected between the sample distribution and uniform distribution

$\rightarrow$ This test is valid only for large sample say, $N \geq 50$.

Example Consider the following sequence of 100 nos

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 0.43 | 0.09 | 0.52 | 0.98 | 0.78 | 0.44 | 0.21 | 0.12 | 0.64 | [illegible] |
| 0.38 | 0.67 | 0.97 | 0.46 | 0.07 | 0.18 | 0.49 | 0.47 | 0.29 | [illegible] |
| 0.69 | 0.99 | 0.77 | 0.76 | 0.65 | 0.14 | 0.25 | 0.37 | 0.99 | [illegible] |
| 0.74 | 0.03 | 0.71 | 0.28 | 0.65 | 0.10 | 0.54 | 0.13 | 0.87 | [illegible] |
| 0.97 | 0.17 | 0.32 | 0.91 | 0.28 | 0.39 | 0.56 | 0.73 | 0.93 | [illegible] |
| 0.99 | 0.71 | 0.99 | 0.64 | 0.50 | 0.66 | 0.01 | 0.24 | 0.81 | [illegible] |
| 0.73 | 0.15 | 0.45 | 0.10 | 0.18 | 0.82 | 0.96 | 0.43 | 0.57 | [illegible] |
| 0.27 | 0.34 | 0.65 | 0.79 | 0.03 | 0.49 | 0.69 | 0.85 | 0.37 | [illegible] |
| 0.60 | 0.93 | 0.48 | 0.42 | 0.04 | 0.46 | 0.04 | 0.91 | 0.97 | [illegible] |
| 0.81 | 0.62 | 0.79 | 0.88 | 0.46 | 0.74 | 0.06 | 0.11 | 0.92 | [illegible] |

Use the chi-square test with $\alpha = 0.05$, to test the hypothesis that the nos are uniformly distributed on the interval $[0,1]$ can be rejected.

Solution 1) Define the hypothesis for testing the uniformity as

$H_0 : R_i \sim U[0,1]$

$H_1 = R_i \nsim U[0,1]$

2) Choose the value of $n$ such that $E_i \geq 5$.

Since $N = 100$, and $E_i = N/n$.

$\therefore 100/n \geq 5 \Rightarrow 100/5 \geq n \Rightarrow n \leq 20$

Let $n = 10$ intervals of equal length, namely $[(0,0.1), (0.2, 0.3), \ldots (0.9, 1.0)]$

3) Compute the test statistics

| Interval | $O_i$ | $E_i = N/n$ | $\frac{(O_i - E_i)^2}{E_i}$ |
|---|---|---|---|
| $(0, 0.1) - 1$ | 8 | 10 | 0.4 |
| $(0.1, 0.2) - 2$ | 9 | 10 | 0.1 |
| $(0.2, 0.3) - 3$ | 10 | 10 | 0.6 |
| $(0.3, 0.4) - 4$ | 6 | 10 | 1.6 |
| $(0.4, 0.5) - 5$ | 13 | 10 | 0.9 |
| $(0.5, 0.6) - 6$ | 8 | 10 | 0.4 |
| $(0.6, 0.7) - 7$ | 11 | 10 | 0.1 |
| $(0.7, 0.8) - 8$ | 12 | 10 | 0.4 |
| $(0.8, 0.9) - 9$ | 7 | 10 | 0.9 |
| $(0.9, 1.0) - 10$ | 16 | 10 | 3.6 |

$$\chi_0^2 = \sum_{i=1}^{n} \frac{(O_i - E_i)^2}{E_i}$$

$= (0.4 + 0.1 + 0.0 + 1.6 + 0.9 + 0.4 + 0.1 + 0.$
$0.9 + 3.6)$
$= 8.4$

→ Determine critical value $\chi^2_{\alpha, n-1}$ for specified significance level $\alpha$ with $n-1$ degree of freedom from table A.6

Since $\alpha = 0.05$, degree of freedom $= 10-1$

$\chi^2_{0.05, 9} = 16.9$

5) Since $\chi_0^2 = 8.4 < \chi^2_{0.05, 9} = 16.9$

$\therefore$ $H_0$ is not rejected

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Candidate Roll No. ____________ (In figures)

Test Exam.

Name : ______________________

Date : ______________ 20

Junior Supervisor's full Signature with Date

Examination : ______________ Branch/Semester ______________

Subject : ______________________

| Question No. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | Total |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Marks Obtained | | | | | | | | | | | | | |

## Test for Independence

1) **Runs Test :-**

→ The runs test analyzes an orderly grouping of nos in a sequence to test the hypothesis of independence.

→ A run is defined as succession of similar events preceded and followed by different event.

→ The length of the run is the number of events that occur in the run.

→ e.g consider the following sequence generated by tossing of coin 10 times

$\underbrace{T\ T}_{2}\ \underbrace{H\ H\ H}_{3}\ \underbrace{T}_{1}\ \underbrace{H\ H}_{2}\ \underbrace{T\ T}_{2}$

→ In above example, there are five runs.

→ The length of each run is 2, 3, 1, 2 and 2.

→ There are two concerns in a runs test namely no of runs and length of runs.

→ Three types of runs test are used namely runs up and runs down, runs above and runs below the mean and length of runs.

→ In all three test the actual values are compared with expected values using the chi square test.

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(1) * Runs up and runs down

→ An up run is a sequence of nos each of which is followed by a larger number.

→ Similarly a down run is a sequence of nos. each of which is followed by smaller no

→ e.g consider following sequence of 10 nos
0.57 0.64 0.34 0.45 0.22 0.01 0.23 0.78 0.88 0.[illegible]

→ The nos are given a "+" or a "−" depending on whether they are followed by a larger no or a smaller number.

→ Since there are 10 nos and all of them are different so there will be 9 +'s and −'s.

→ The last no is followed by "no event" and hence it will not get neither + nor a −

→ The sequence of 9 +'s and −'s is given.
+ − + − − + + + −

→ Here each succession of +'s and −'s form a run.

→ In above example there are six runs and length of each run is 1, 1, 1, 2, 3 and 1.

→ There can be a situation of few runs (only one run) or too many runs (N−1, if N is total no of observations).

→ Both are unlikely case for a valid random generator.

→ The more likely case is that the no of runs will be somewhere betw these two extremes.

→ Let 'a' be the total no of runs found in the sequence.

→ For N > 20, 'a' is approximated by a normal distribution $N(\mu_a, \sigma_a^2)$ which can be used to test the independence of nos from a generator.

→ Finally a standardize is used to test the independence on the basis of runs up and runs down in given sequence of nos.

Algorithm

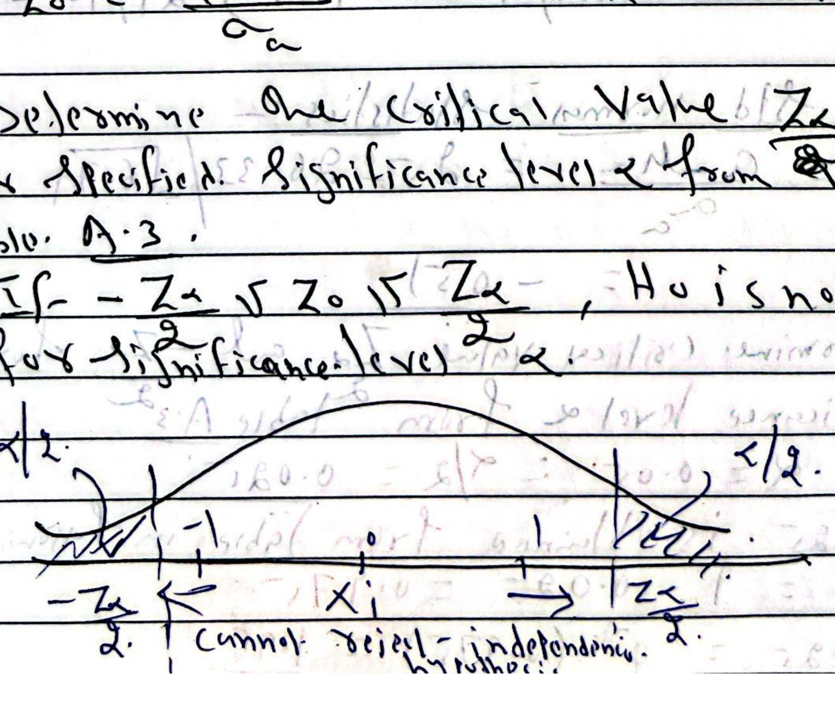
1) Define hypothesis for testing.
   $H_0$ : $R_i \sim$ independently.
   $H_1$ : $R_i \nsim$ independently.
2) write down sequence of runs up and down.
3) Count the total no of runs (a), Present in sequence.
4) compute mean and variance of a.

$$\mu_a = \frac{2N-1}{3} \text{ and } \sigma_a^2 = \frac{16N-29}{90}$$

5) compute std normal statistics.

$$Z_0 = \frac{a-\mu_a}{\sigma_a} \text{ where } Z_0 \sim N(0,1)$$

6) Determine the critical value $Z_{\frac{\alpha}{2}}$ and $-Z_{\frac{\alpha}{2}}$ for specified significance level $\alpha$ from table A.3.
7) If $-Z_{\frac{\alpha}{2}} \le Z_0 \le Z_{\frac{\alpha}{2}}$, $H_0$ is not rejected for significance level $\alpha$.



7) Since $-Z_{0.025} = -1.96 \le Z_0 = -0.51 \le Z_{0.025} = 1.96$.
$\therefore$ $H_0$ is not rejected.

Example:-
Consider the following sequence of 40 numbers.

0.52 0.99 0.46 0.58 0.64 0.25 0.48 0.11 0.20
0.97 0.44 0.43 0.94 0.82 0.60 0.73 0.69 0.?
0.04 0.81 0.85 0.30 0.47 0.96 0.17 0.72 0.6?
0.10 0.60 0.34 0.65 0.79 0.44 0.02 0.37 0.?

Based on runs up and runs down determine whether the hypothesis of independence can be rejected, where $\alpha = 0.05$.

$\rightarrow$ Solution 1) Define hypothesis.
$H_0$: $R_i \sim$ independently.
$H_1$: $R_i \nsim$ independently.

2) The sequence of runs up and down is.
+ − + + − + − + − +
− − + − − + − − + +
+ + − + + − + − − −
− + − + + − − + + +

3) The total number of runs $a = 25$

4) Mean and variance of 'a' is.
$\mu_a = 2N-1/3 = 2(40)-1/3 = 26.33$
$\sigma_a^2 = 16N-29/90 = 16(40)-29/90 = 6.79$

5) The std normal statistics:-
$Z_0 = \frac{a-\mu_a}{\sigma_a} = 25 - 26.33/\sqrt{6.79}$
$= -0.51$

6) Determine critical value $Z_{\alpha/2}$ and $-Z_{\alpha/2}$ for specified significance level $\alpha$ from table A.3
Since $\alpha = 0.05$ $\therefore \alpha/2 = 0.025$
Now $Z_{0.025}$ is obtained from table in following manner
$\phi(Z_{0.025}) = 1 - 0.025 = 0.975$
$Z_{0.025} = \phi^{-1}(0.975) = 1.96$

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Candidate Roll No. ______ (In figures)

**Test Exam.**

Name: ______

Date: ______ 20

Examination: ______ Branch/Semester ______

Subject: ______

| Junior Supervisor's full Signature with Date |
|---|
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| Question No. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | Total |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Marks Obtained | | | | | | | | | | | | | |

(2) Runs above and below the mean

→ The test for runs up and runs down is not completely adequate to evaluate the independence of sequence of numbers.

→ So runs are redefined as being above the mean or below the mean.

→ A "+" sign will be used to denote a no above the mean and "−" sign will denote a no below mean.

→ consider following sequence of 10 nos

0.57 0.64 0.34 0.45 0.22 0.01 0.23 0.78 0.88 0.19

→ The nos are given a "+" or a "−" depending on whether they are greater than or smaller than the expected mean which is $(0+0.99)/2 = 0.495$

→ The sequence of 10 +'s and −'s is given below.

+ + − − − − − + + −

→ In above example there are four runs.

→ The length of each run is 2, 5, 2 and 1.

→ Let $n_1$ and $n_2$ be the no of observations above and below mean respectively.

→ The maximum no of runs is $N = n_1 + n_2$ whereas minimum is one.

→ Let 'b' is total no of runs in a sequence.

→ For either $n_1 > 20$ or $n_2 > 20$, b is approximated by a normal distribⁿ $N(\mu_b, \sigma_b^2)$ which can be used to test the independence of nos from a generator.

→ Finally Std. normal test statistics $Z_0$ is developed and it is compared with the critical value.

→ The following algorithm is used to test the independence on basis of runs above and below mean in the given sequence of nos.

Algorithm :-

1) Define hypothesis for testing the independence.

$H_0$ : $R_i \sim$ independently

$H_1$ : $R_i \nsim$ independently

2) Write down sequence of runs above and runs below the mean.

3) Count the no of observations above mean ($n_1$), no of observations below mean ($n_2$) and total no of runs ($b$) present in the sequence.

4) Compute mean and variance of $b$.

$$\mu_b = \frac{2n_1n_2}{N} + \frac{1}{2} \quad \text{and} \quad \sigma_b^2 = \frac{2n_1n_2(2n_1n_2 - N)}{N^2(N-1)}$$

5) Compute std normal statistics.

$$Z_0 = \frac{b-\mu_b}{\sigma_b} \quad \text{where } Z_0 \sim N(0,1).$$

6) Determine the critical value $Z_{\frac{\alpha}{2}}$ and $-Z_{\frac{\alpha}{2}}$ for specified significance level $\alpha$ from table.

7) If $-Z_{\frac{\alpha}{2}} \leq Z_0 \leq Z_{\frac{\alpha}{2}}$ ∴ $H_0$ is not rejected for the significance level $\alpha$.

Example consider following sequence of 40 nos

0.09 0.41 0.23 0.68 0.89 0.72 0.12 0.45 0.08 0.32
0.53 0.13 0.65 0.97 0.14 0.49 0.55 0.46 0.77 0.28
0.81 0.63 0.40 0.57 0.02 0.16 0.33 0.86 0.99 0.22
0.76 0.48 0.61 0.39 0.43 0.78 0.20 0.35 0.17 0.93.

Determine whether there is an excessive no of runs above or below the mean. Use $\alpha = 0.05$

⟹ 1) Define hypothesis for testing indetendence as.

$H_0$ : $R_i \sim$ independently.

$H_1$ : $R_i \not\sim$ independently.

2) The sequence of runs above and below the mean (0.495) is.

− − − + + + − − − −
+ − + + − − + − + −
+ + − + − − − + + −
+ − + − − + − − − +.

3) The no of observn above mean $n_1 = 17$ and $n_2 = 23$ → below mean.

Total no of runs. $b = 24$

4) Mean and variance of $b$

$\mu_b = \frac{2n_1n_2}{N} + \frac{1}{2} = \frac{2(17)(23)}{40} + \frac{1}{2} = 20.05$

$\sigma_b^2 = \frac{2n_1n_2(2n_1n_2 - N)}{N^2(N-1)} = \frac{2(17)(23)(2(17)(23)-40)}{40^2(40-1)}$

$= 9.3$

5) The std normal statistics

$Z_0 = \frac{b - \mu_b}{\sigma_b} = 24 - 20.05 / \sqrt{9.3} = 1.295$

6) Since $\alpha = 0.05$ $\therefore \alpha/2 = 0.025$

$\therefore Z_{0.025} = 1.96$

7) Since $-Z_{0.025} = -1.96 \leq Z_0 = 1.295 \leq Z_{0.025} = 1.96$. ∴ $H_0$ is not rejected.

$D = \text{Max} |F(x) - S_N(x)|$

$= \underline{0.0348}$

5) Determine the critical value, $D_\alpha$ for specified value of significance level $\alpha$ and the sample size N from table A.8. Since $\alpha = 0.05$ and $N = 110$

$\therefore D_{0.05} = \dfrac{1.36}{\sqrt{110}}$

$= \underline{0.1297}$

6) Since $D = 0.0348 < D_{0.05} = 0.1297$

$\therefore$ $H_0$ is not rejected.

## * Poker Test :- *

→ It tests the freq of certain digits in a series of numbers.

→ Here we show only the three digit version for testing the independence property.

→ In this case the generated random numbers are rounded to three digits. Then each random number is classified into one of following categories.

→ 1) All digits are different from each other.

→ 2) All digits are identical.

→ 3) There is exactly one pair of identical digits

The focus of test is not the sequence from one random number to another. Rather this test focuses on the internal digits within a given number.

(In figures)

**Test Exam.**

Name:

Date: 20

Examination: Branch/Semester

Subject:

| Junior Supervisor's full Signature with Date |
|---|
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| Question No. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | Total |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Marks Obtained | | | | | | | | | | | | | |

→ If digits are truely generated randomly, the following results hold.

1) The individual numbers can all be different and which has a probability.

$P = P(2^{nd}$ digit diff from $1^{st}) \; P(3^{rd}$ digit diff from $1^{st}, 2^{nd})$

$= (0.9)(0.8) = \underline{0.72}$

2). The individual numbers can all be same and which has a probability.

$P = P(2^{nd}$ digit same as $1^{st}) \; P(3^{rd}$ same as $1^{st})$

$= (0.1)(0.1) = \underline{0.01}$

3) There can be one pair of like digits and which has a probability.

$P = 1 - [P($three diff digits$) + P($three like digits$)]$

$= 1 - (0.72 + 0.01) = \underline{0.27}$

→ The observed value is then compared with expected value using the <u>chi square test</u>.

Algorithm

→ 1) <u>Define</u> <u>hypothesis</u> <u>for</u> <u>testing</u> <u>independence</u>.

→ 2) Generate freq distrib table for above three combination and apply <u>chi square</u> <u>test</u>.

| Combination, $i$ | Observed Freq, $O_i$ | Expected Freq $E_i = P \times N$ | $\frac{(O_i - E_i)^2}{E_i}$ |
|---|---|---|---|

3) Compute the sample test statistics.

$$\chi_0^2 = \sum_{i=1}^{n} \frac{(O_i - E_i)^2}{E_i}$$

4) Determine the critical value for specified significance level $\alpha$ with $n-1$ degrees of freedom from table A.6

5) If $\chi_0^2 > \chi^2_{\alpha, n-1} \Rightarrow H_0$ is rejected, else no difference has been detected betwn the sample distribution and uniform distribution.

**Example** A sequence of 1000 three digit numbers has been generated and an analysis indicates that 560 have three different digits, 380 contain exactly one pair of like digits and 60 contain three like digits. Based on poker test, test whether these nos are independent. Use $\alpha = 0.05$

$\Rightarrow$ 1) Define hypothesis.

2) Generate freq distribn table and apply chi square test

| Combination, i | Observed Freqn $O_i$ | Expected freq $E_i = P \times N$ | $\frac{(O_i - E_i)^2}{E_i}$ |
|---|---|---|---|
| Three diff digits, 1 | 560 | 720 | 35.56 |
| Three like digits, 2 | 60 | 10 | 250.00 |
| Exactly one pair, 3 | 380 | 270 | 44.82 |

3) The sample test statistics

$$\chi_0^2 = \sum_{i=1}^{n} \frac{(O_i - E_i)^2}{E_i}$$

$$= (35.56 + 250.00 + 44.82) = 330.38$$

4) Determine the critical value for specific significance level $\alpha$ and $n-1$ degree of freedom from table A.6

Since $\alpha = 0.05$, $n-1 = 3-1 = 2$

$\therefore \chi^2_{0.05, 2} = \underline{5.99}$

5) Since $\chi_0^2 = 330.38 > \chi^2_{0.05, 2} = \underline{5.99}$

$\therefore$ $\underline{\text{Ho is rejected}}$